**This mobile application teaches** children **waste segregation** through **fun, interactive gameplay**. Players sort different types of waste into the correct bins, helping **build quick recognition and muscle memory**. The app uses a **child-friendly interface** with a playful cat mascot that guides and engages users throughout the experience.

**Project :** Kids Learning App

**Platform :** Education

**By :** Alisha Khare and Navya Grover

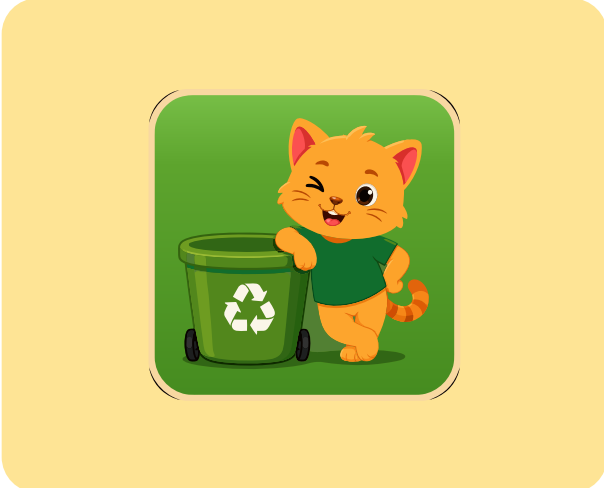
### Learning Made Easy

The experience is structured through a level-based progression system:

- Each level increases in complexity with new waste types, obstacles, and bin variations
- Instant visual and audio feedback reinforces learning

### For Young Minds

Designed with playful characters and child-friendly interactions to encourage eco-awareness and build sustainable habits early.



9:41
Trash Bash
MON
10
Calendar
Photos
Reminders
Camera
Mail
Notes
Cloc

## Insights

**why so difficult?**
Lack awareness about proper waste segregation from early age.

**no muscle memory**
Waste categories are often confusing and hard to remember
Traditional learning methods feel boring and not engaging.

**what happens after?**
Improper segregation at source leads to mixed waste disposal

### What can be done?

- Build muscle memory through repeated gameplay
- Encourage quick identification of waste categories
- Teach segregation at source in an interactive way
- Improve decision-making through visual learning
- Make environmental education engaging and memorable

## But why **gamification**?

- Interactive gameplay improves engagement and attention
- Repetition helps build long-term muscle memory
- Rewards and feedback motivate continuous learning
- Visual learning makes waste identification easier for children
- Learning through play increases retention and habit formation

## 'Design is not just screens'

### Double Diamond Framework

Discover: Identified travel-related pet care trust gaps through user insights. Define: Framed problem of unreliable stays. Develop: Explored unified platform concepts. Deliver: Designed Pawzo with verified sitters, real-time updates, and seamless booking experience.

**Discover**
Brainstorming | Survey | User insights |

**Define**
Empathy Mapping | Pain Points |

**Develop**
Visual Identity | Brand Mark | Icons |

**Deliver**
Wireframe | Design System | Final Design

## User Insights

Waste segregation confusing and difficult to remember

Traditional awareness methods
"BORING,"
to learn

Less Learning through action

LEARNING EXPERIENCE

Interactive = Builds Habit

### User Persona

"A working parent who wants her child to develop responsible habits from an early age."

**Priya Sharma**
(Working Parent)

**Age:** 36
**Location:** Delhi, India
**Primary Goal:**
Encourage environmental responsibility in her child.

- Interactive and engaging learning experiences
- Simple and visual explanations for children
- A fun way to encourage habit formation through daily interaction

### Empathy Mapping

**Says**

- "My child doesn't understand waste segregation properly."
- "Kids forget these habits very quickly."
- "Environmental awareness should start early."

**Thinks**

- Waste management concepts are difficult for children to grasp
- Traditional teaching methods are not engaging enough
- Children need more practical exposure to learn habits

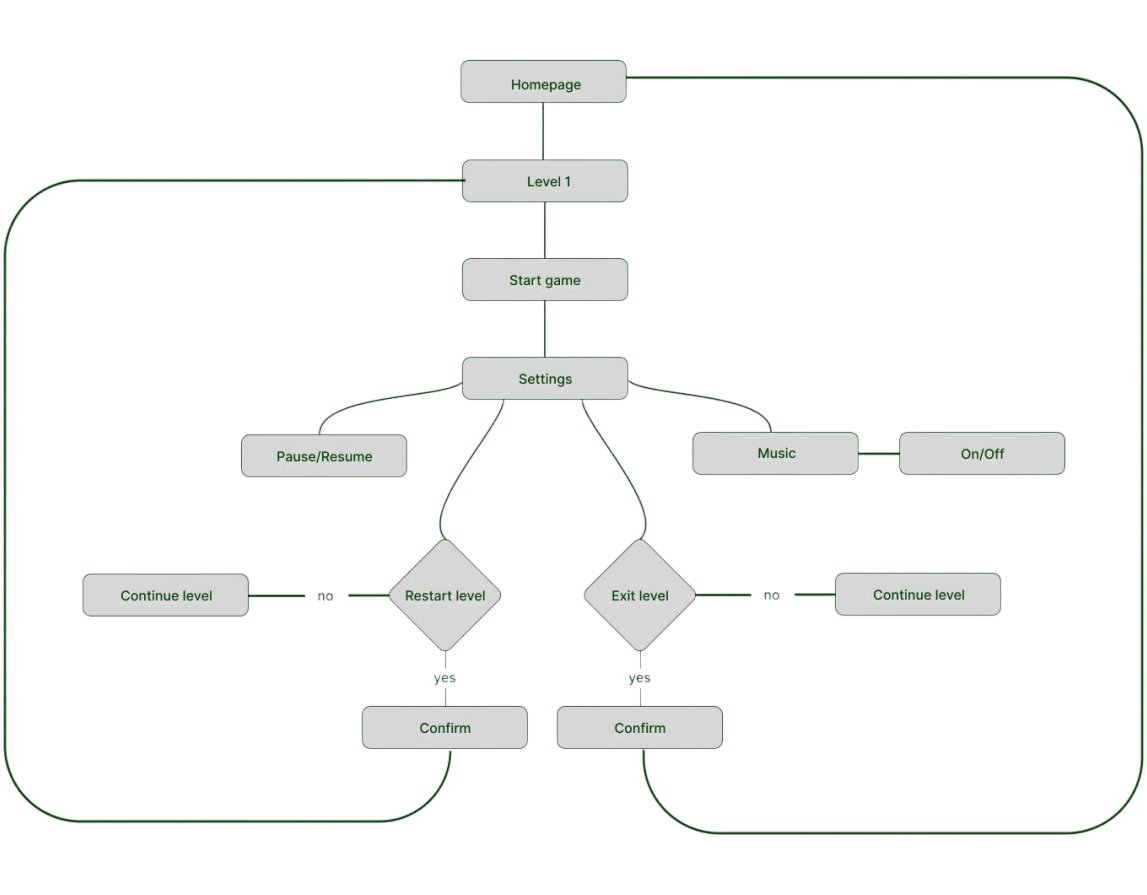
**Does**

- Reminds children to throw waste properly
- Tries teaching cleanliness and recycling habits at home
- Encourages responsible behavior in daily activities

**Feels**

- Concerned about growing environmental issues
- Frustrated by careless disposal habits
- Worried that sustainability is not taught effectively at a young age

**A simple intutive flow** designed to guide kids from home screen to learning activities in just a tap.


Homepage
Level 1
Start game
Settings
Pause/Resume
Music
On/Off
Continue level
no
Restart level
yes
Confirm
Exit level
no
Continue level
yes
Confirm

## Style Guide

A **nature-inspired palette** with greens signify freshness and growth, while browns ground the design in earthiness and responsibility.

Primary Colour
#0A8941

Accent Colour
#F6B004

Accent Colour
#77360C

Accent Colour
#CB7927

Background Colour
#E1FFD8

Text
#000000

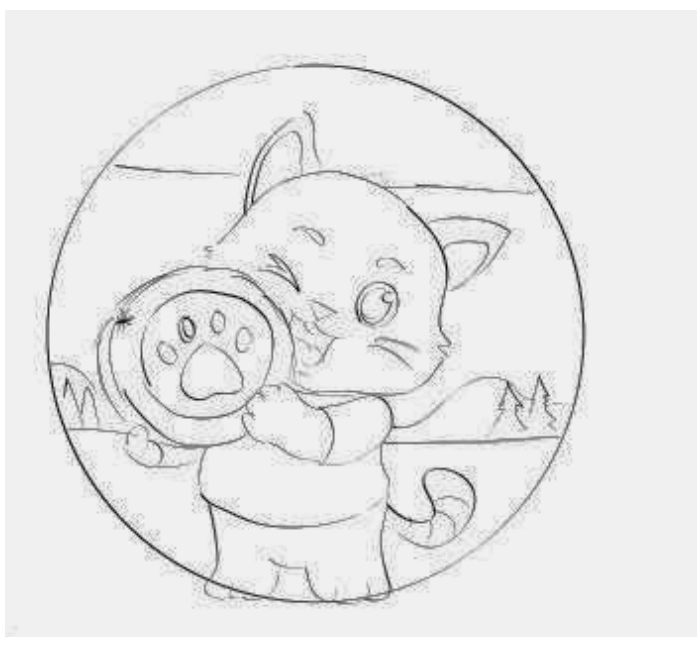
### Typeface

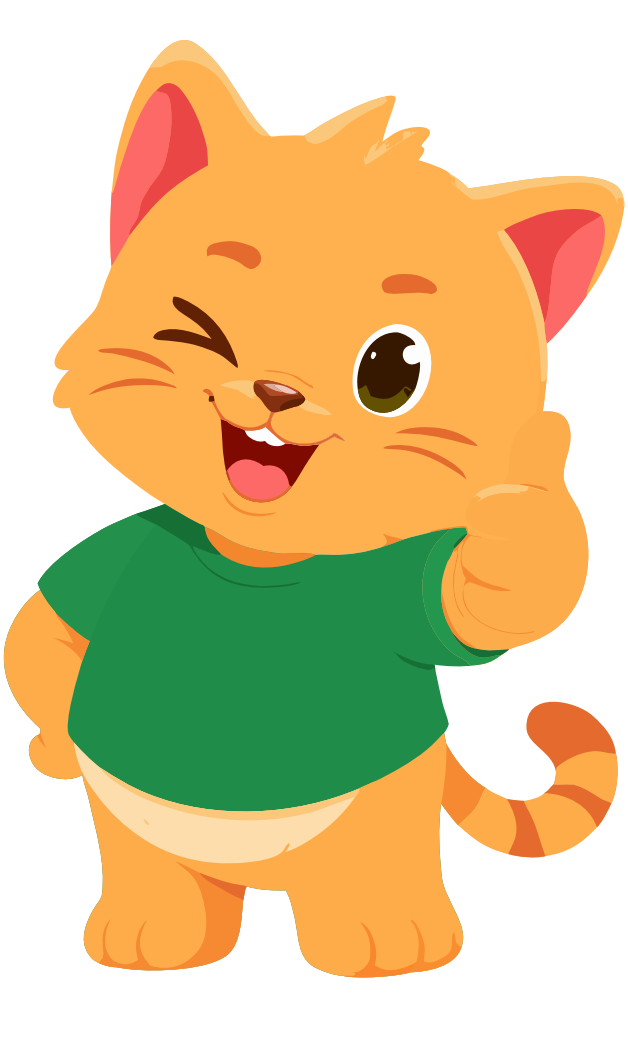
FREDOKA

Bold, SemiBold, Medium, Regular, Light

AaBbCcDd AaBbCcDd

123456!?- 123456!?-

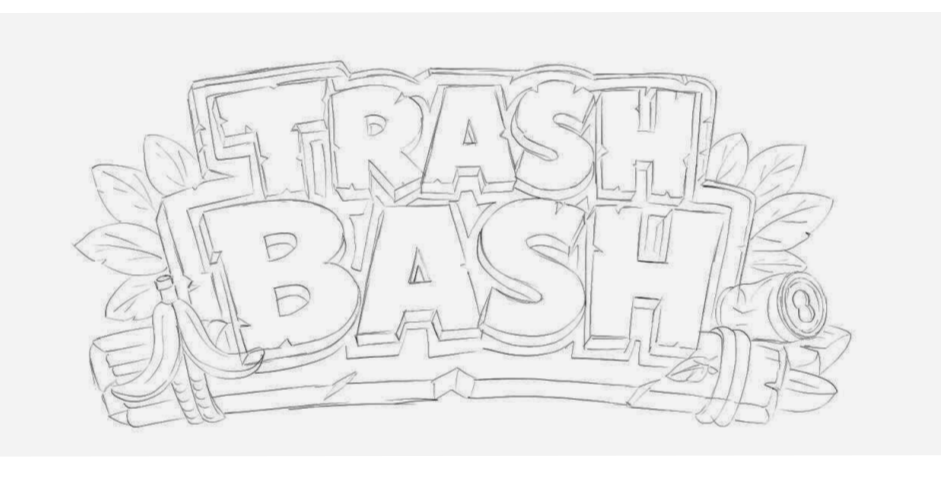


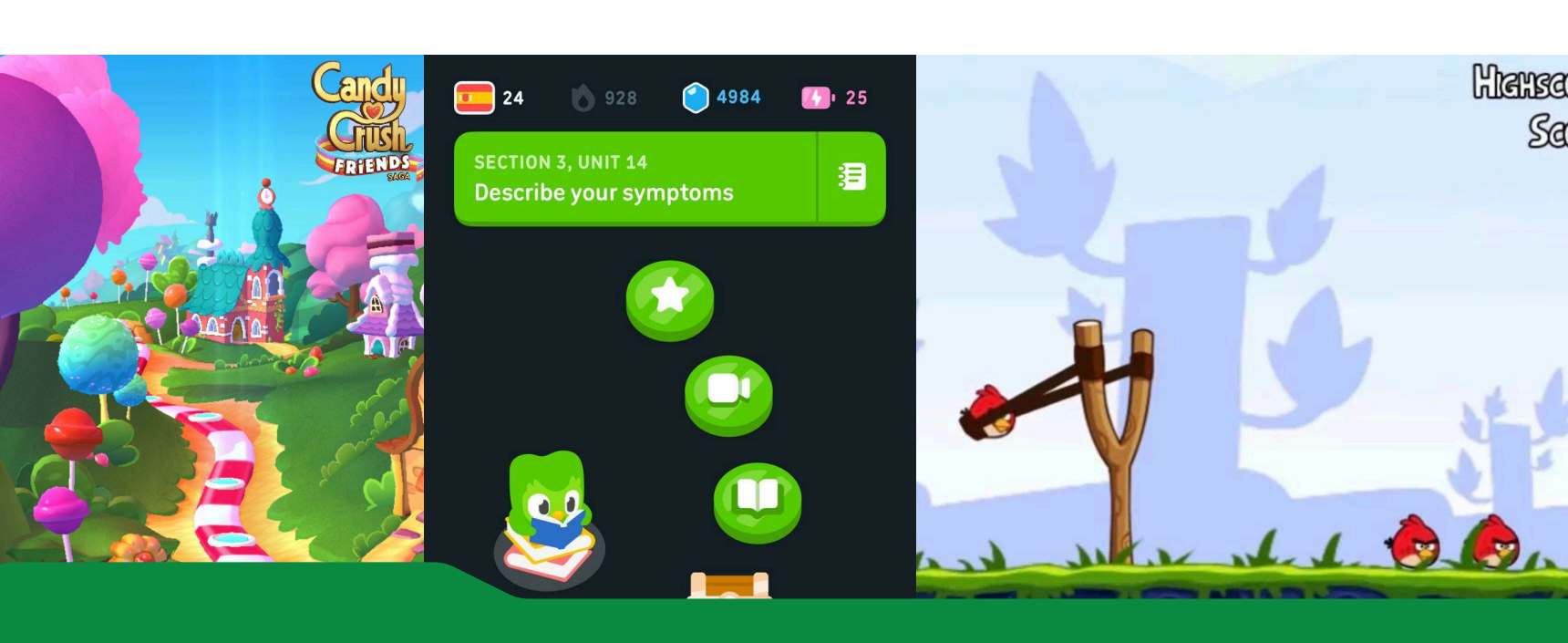


**CAT AS A MASCOT**

Cats are naturally clean animals. They groom themselves regularly and maintain tidy habits. Using a cat as the mascot subtly reinforces the idea of cleanliness and hygiene, making the concept of waste segregation more relatable and easy for children to understand.

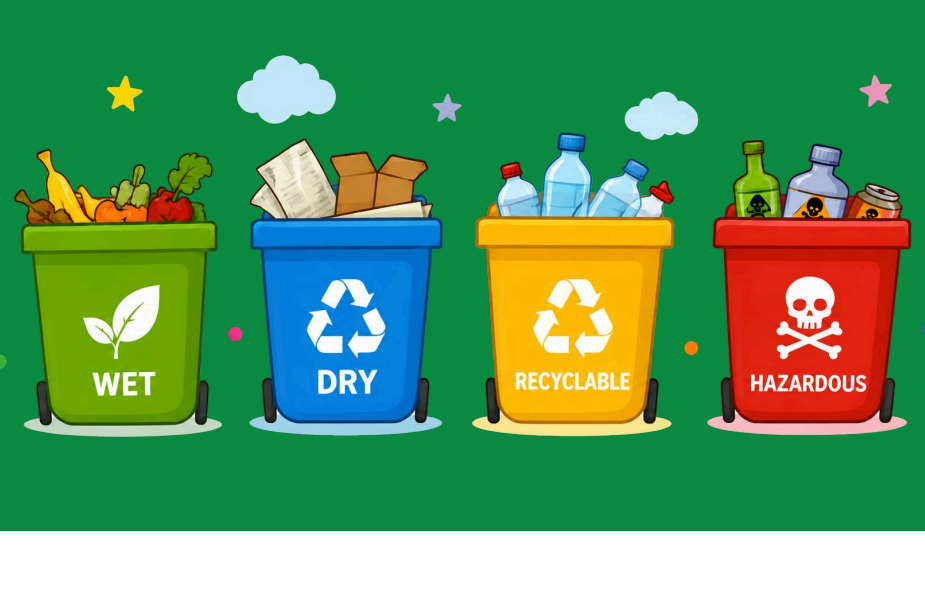
### Logo




Candy Crush FRIENDS
24
928
4984
25
SECTION 3, UNIT 14
Describe your symptoms
Highsc
Sc

**INSPIRATION AREA**

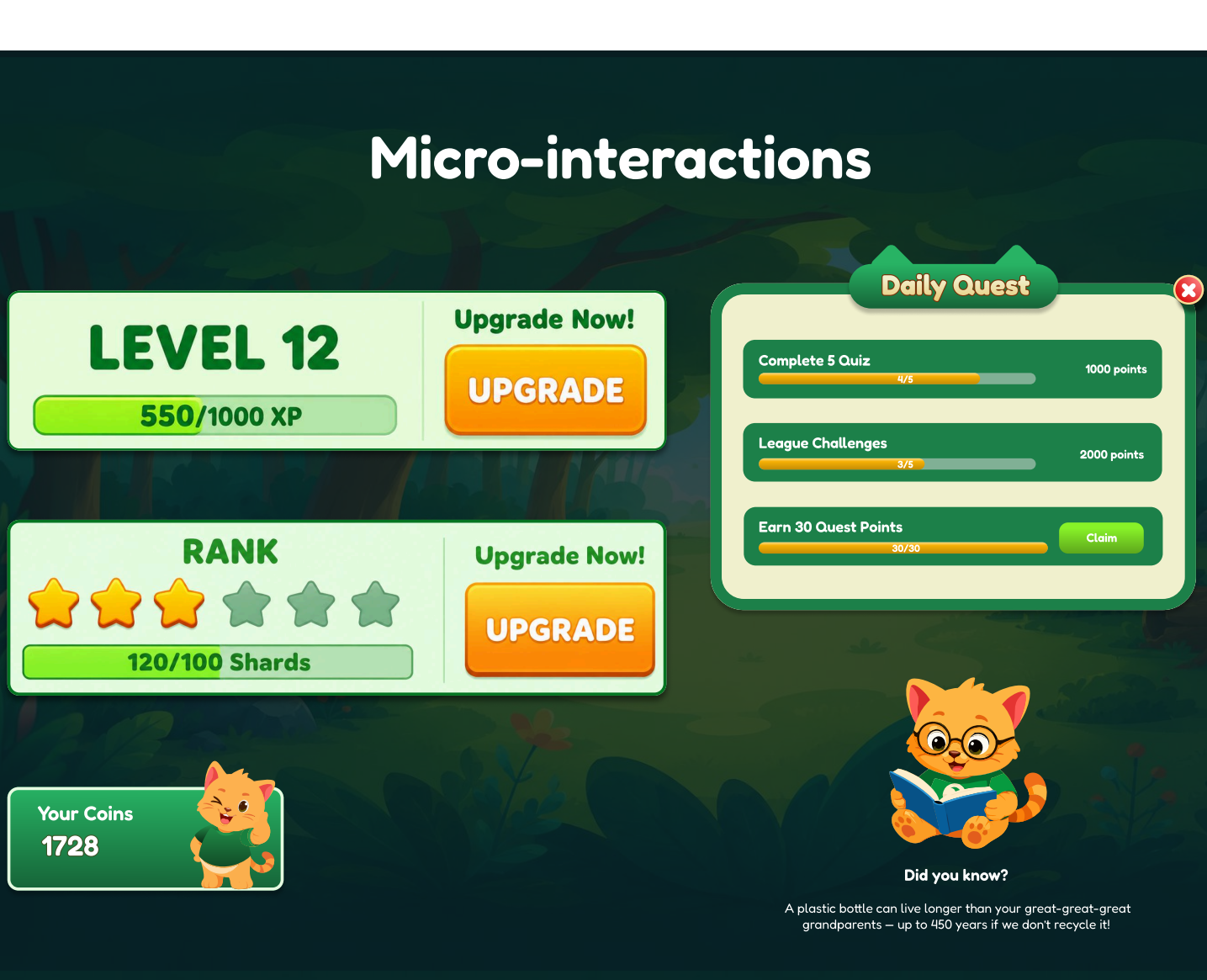
The app takes inspiration from Angry Birds, Duolingo, and Candy Crush. Angry Birds inspired the slingshot-style gameplay, Duolingo influenced the quiz-based habit learning system, and Candy Crush inspired the colorful visuals, rewards, and engaging level progression.


WET
DRY
RECYCLABLE
HAZARDOUS

kid friendly

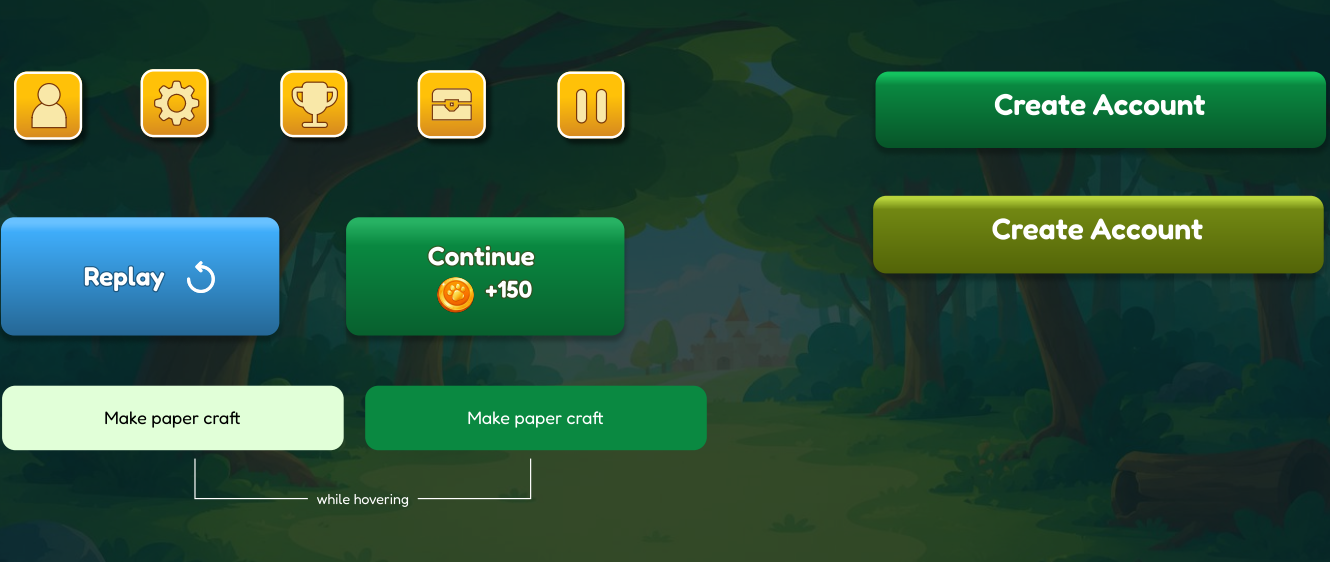
## Design System

### Micro-interactions

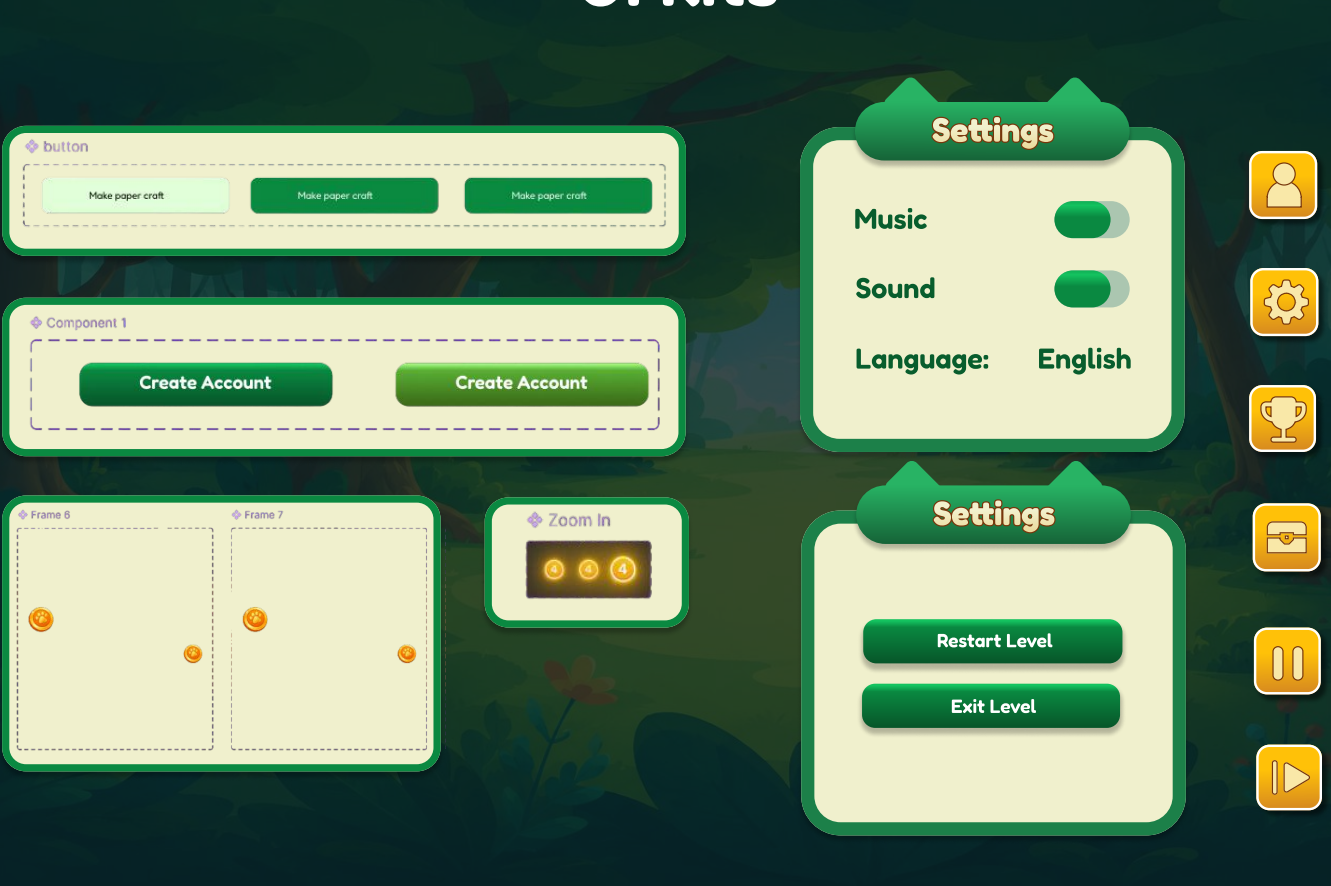

LEVEL 12
550/1000 XP
Upgrade Now!
UPGRADE
RANK
120/100 Shards
Upgrade Now!
UPGRADE
Your Coins
1728
Daily Quest
Complete 5 Quiz
1000 points
League Challenges
2000 points
Earn 30 Quest Points
Claim

Did you know?
A plastic bottle can live longer than your great-great-great grandparents — up to 450 years if we don't recycle it!

### Clear CTAs for Primary Actions


Create Account
Create Account
Replay
Continue
+150
Make paper craft
Make paper craft
while hovering

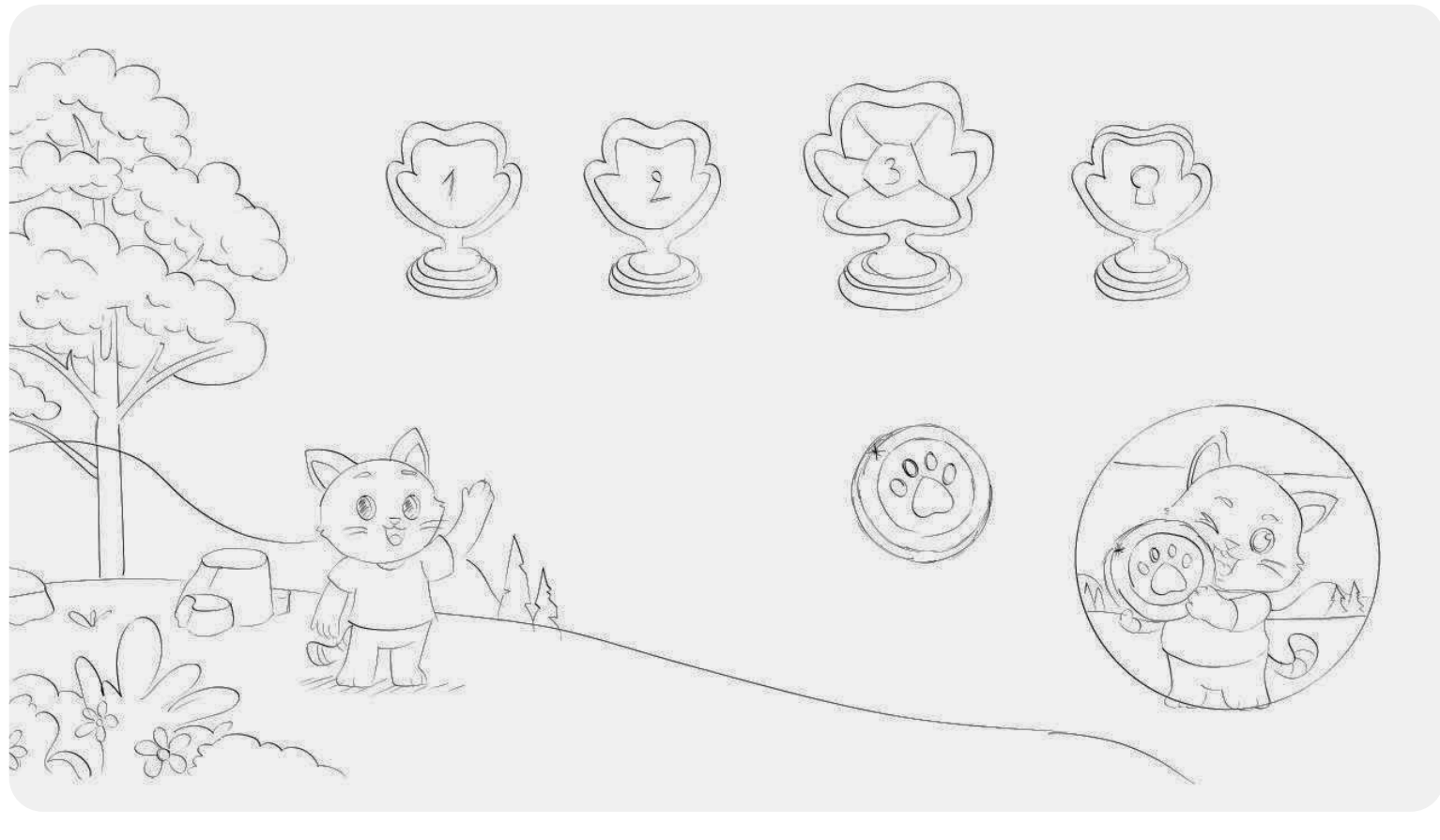
### UI Kits


button
Make paper craft
Make paper craft
Make paper craft
Component 1
Create Account
Create Account
Frame 6
Frame 7
Zoom In
Settings
Music
Sound
Language:
English
Settings
Restart Level
Exit Level

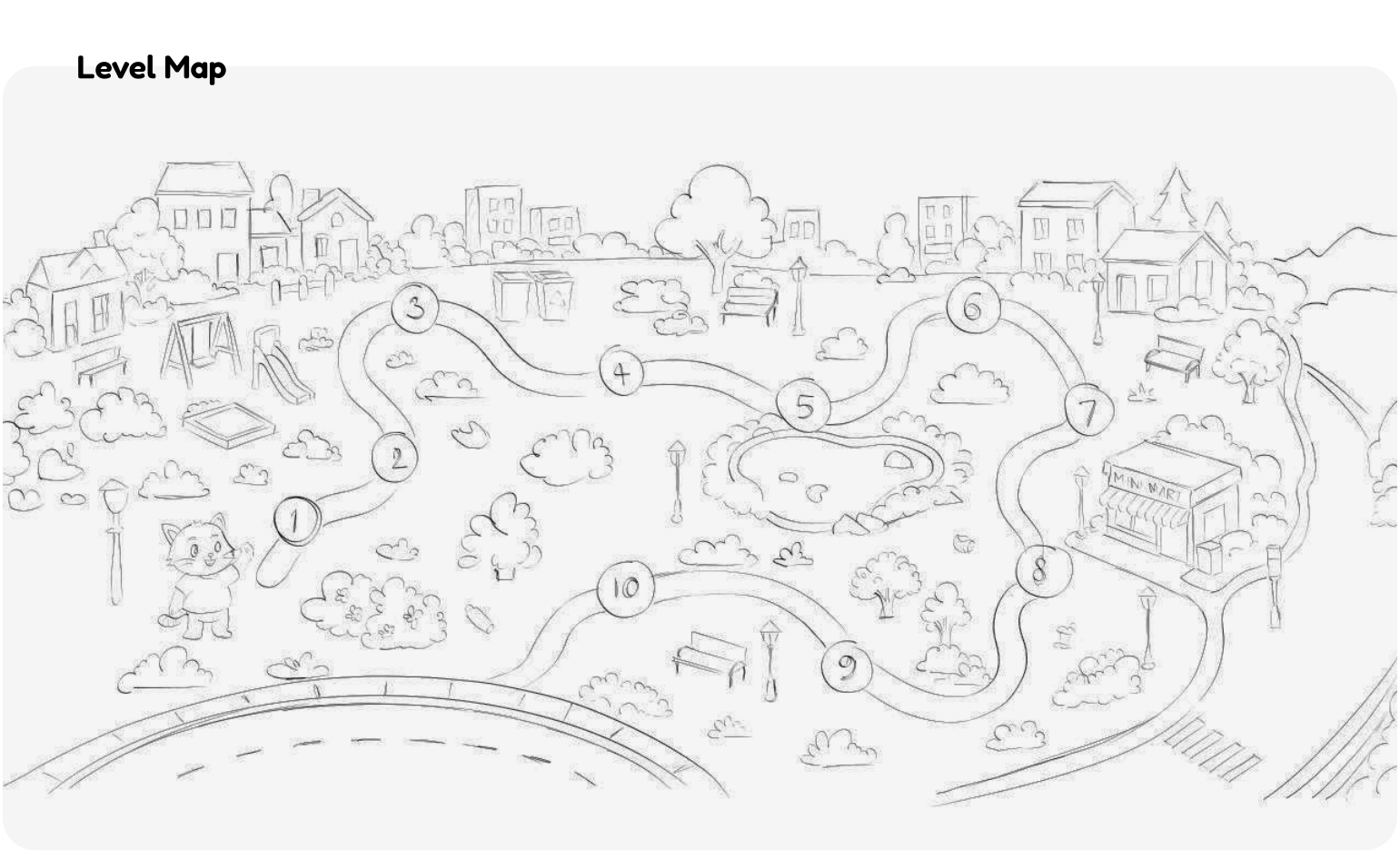
> "When learning feels fun, you naturally grow without even realising it."

## CONCEPT ILLUSTRATIONS

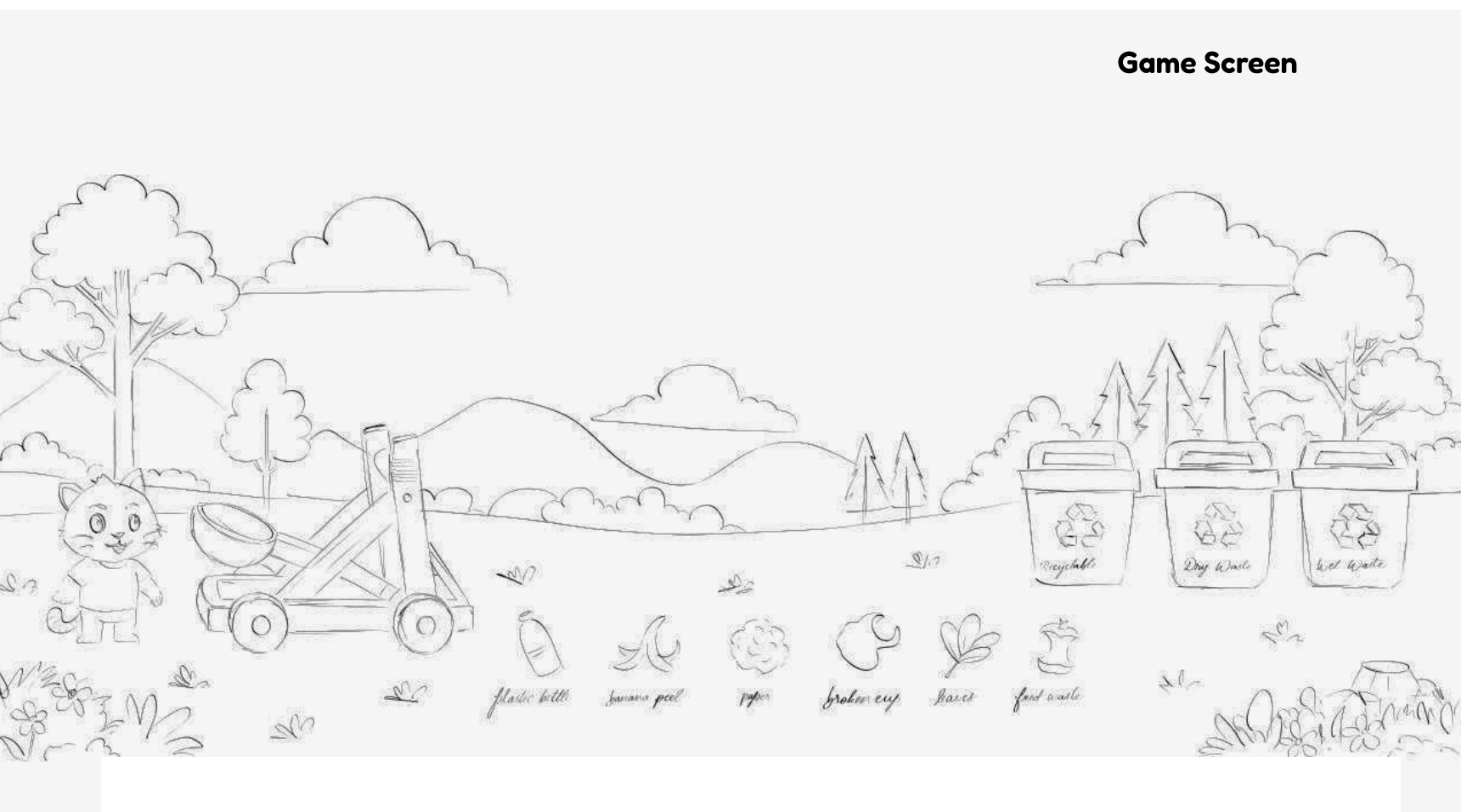
Onboarding Experience



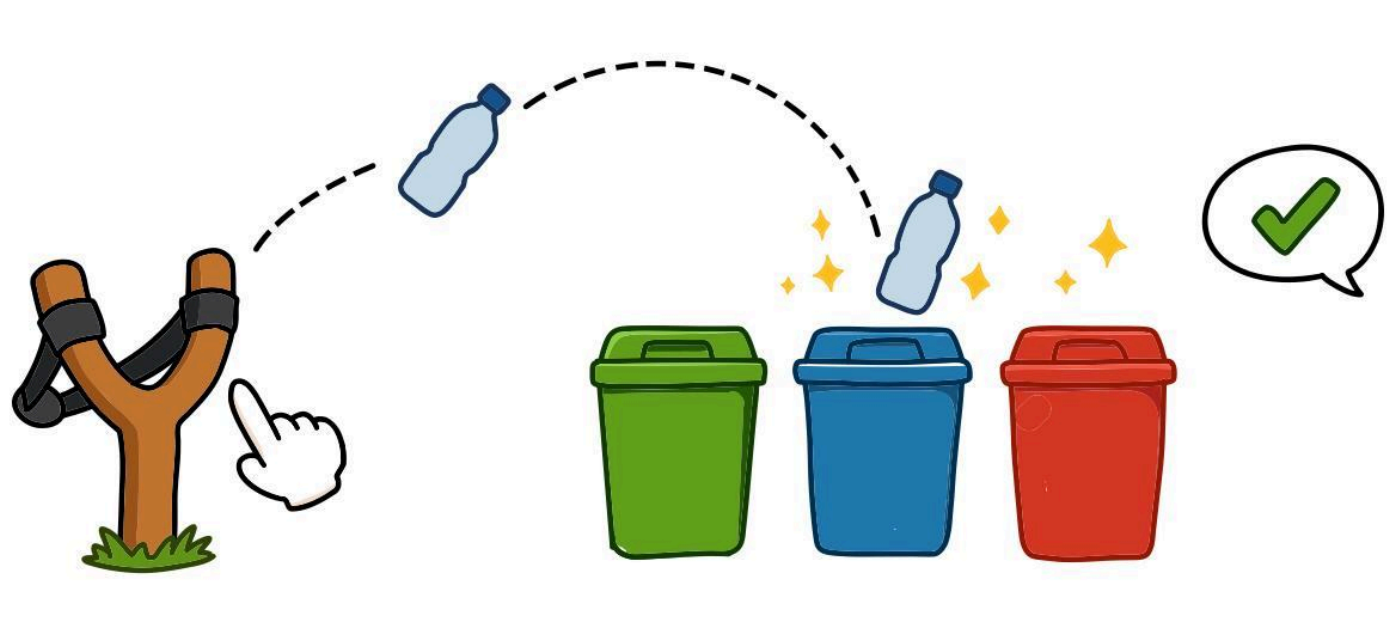
Level Map



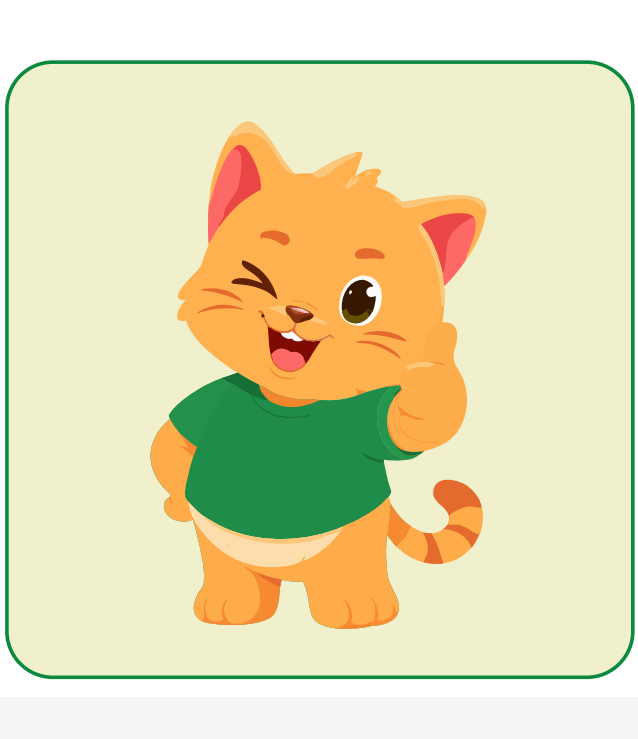
Game Screen

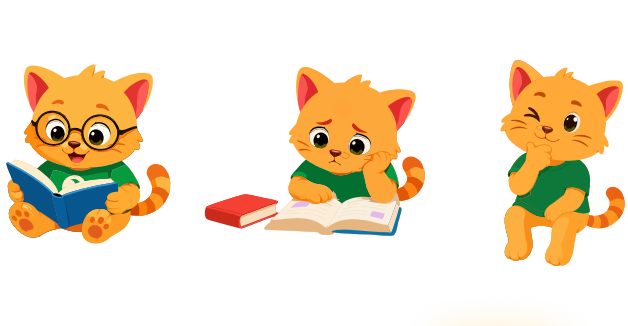

Recyclable
Dry Waste
Wet Waste
plastic bottle
banana peel
paper
broken cup
leaves
food waste

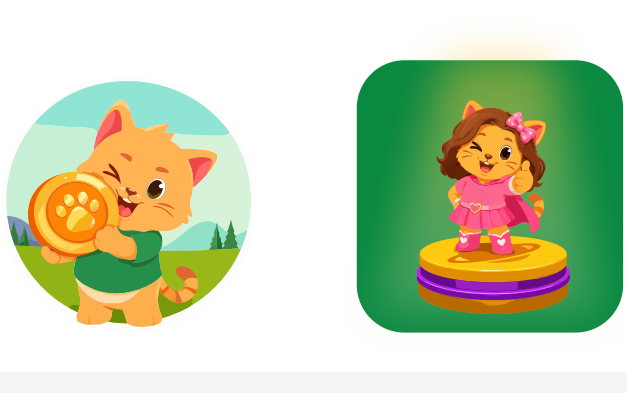
How to play

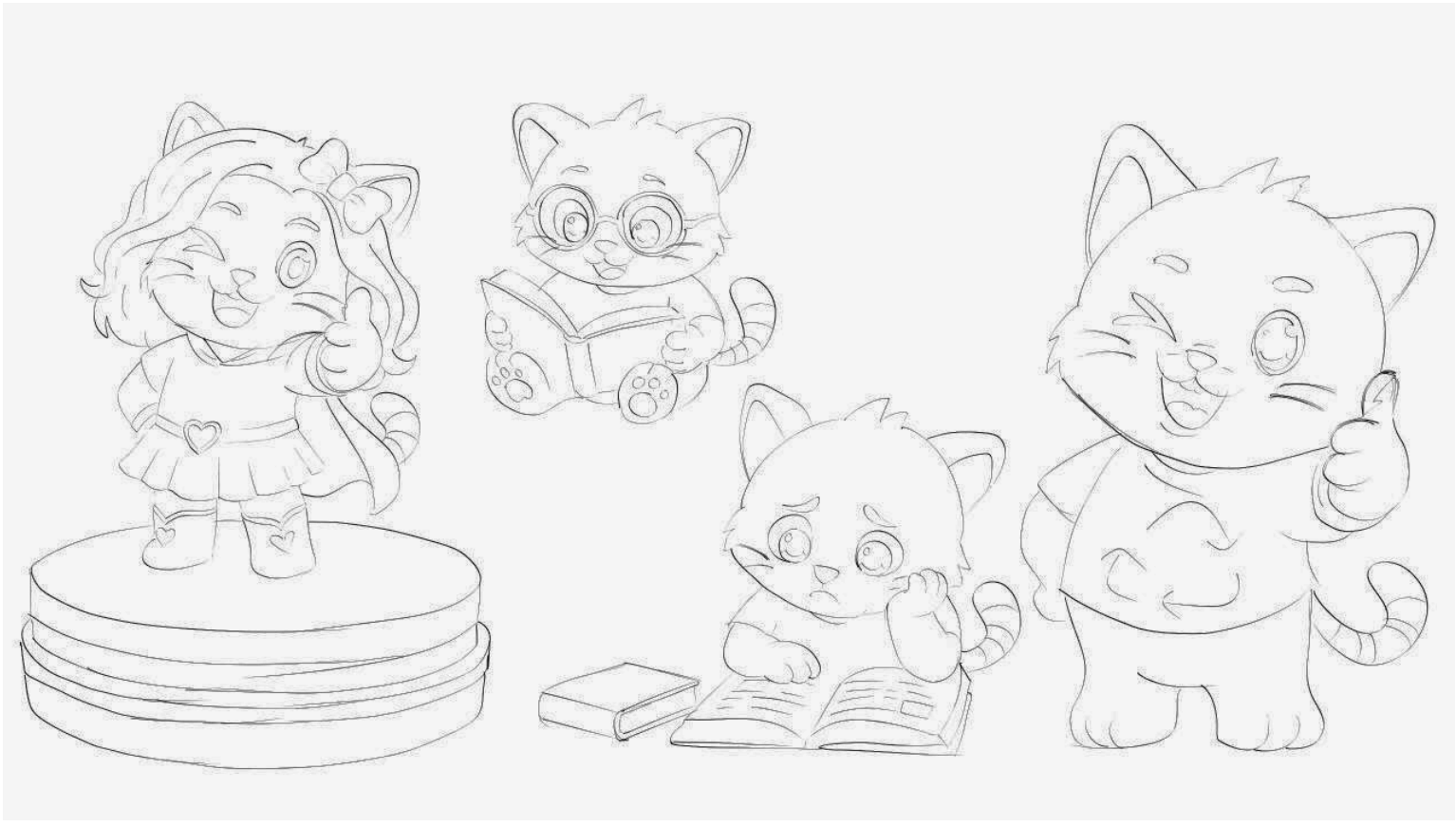


## CHARACTER ILLUSTRATIONS



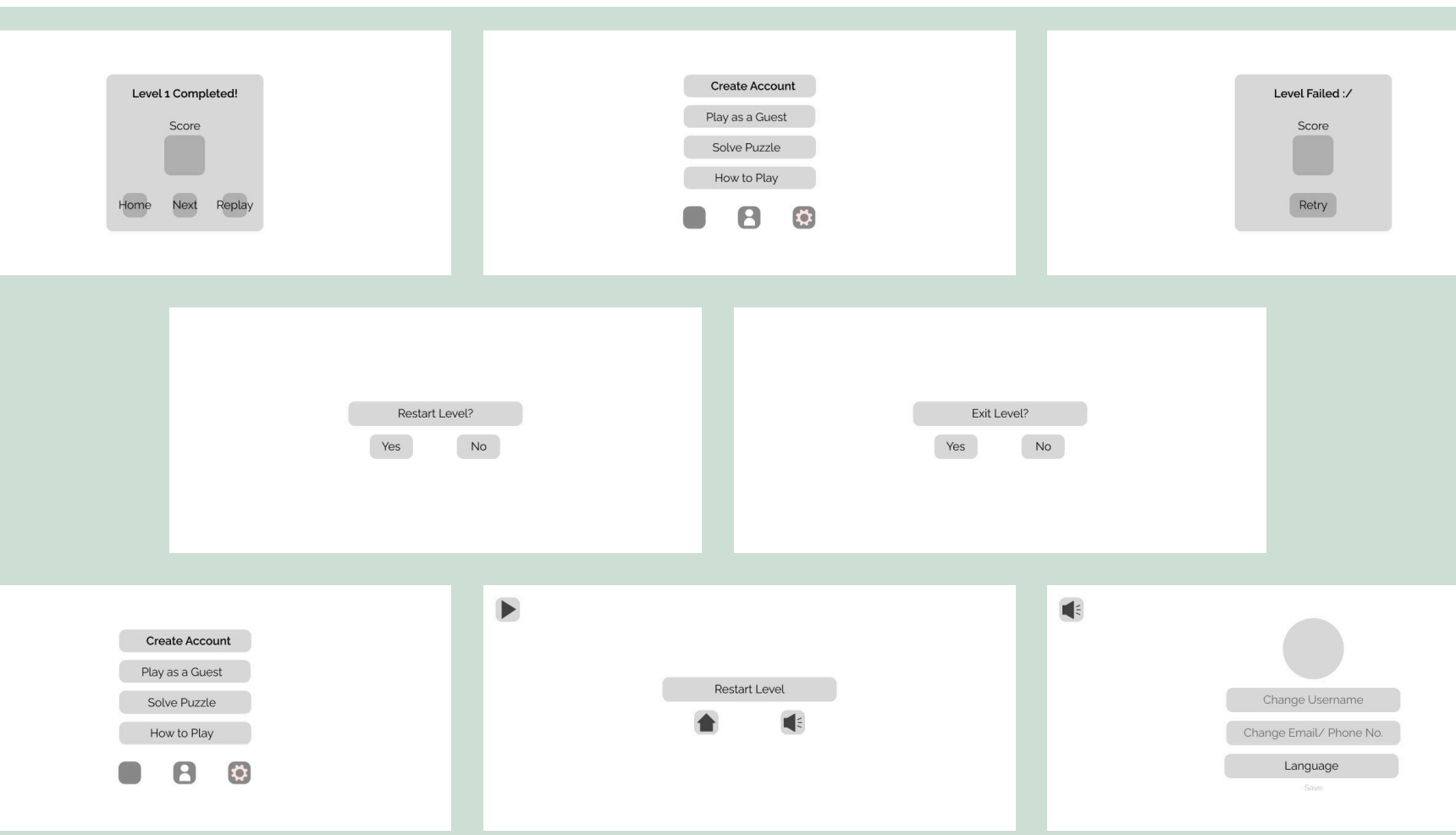






flows

## Wireframes

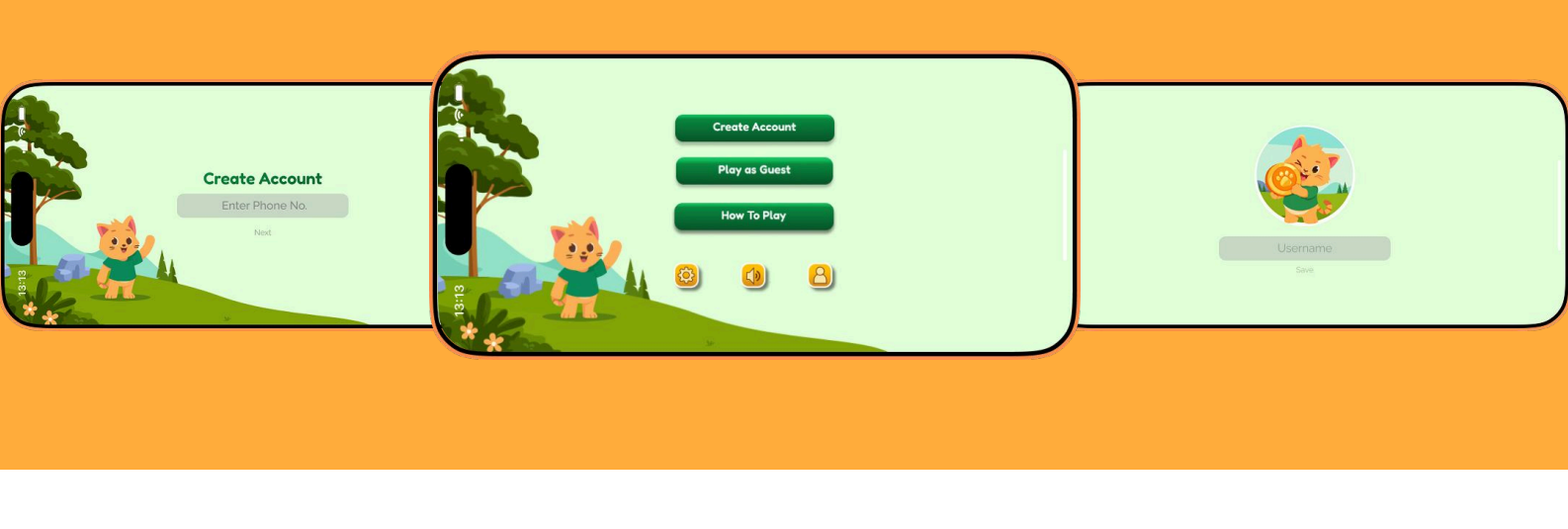

Level 1 Completed!
Score
Home
Next
Replay
Create Account
Play as a Guest
Solve Puzzle
How to Play
Level Failed :/
Score
Retry
Restart Level?
Yes
No
Exit Level?
Yes
No
Create Account
Play as a Guest
Solve Puzzle
How to Play
Restart Level
Change Username
Change Email/ Phone No.
Language

flows

## App Experience

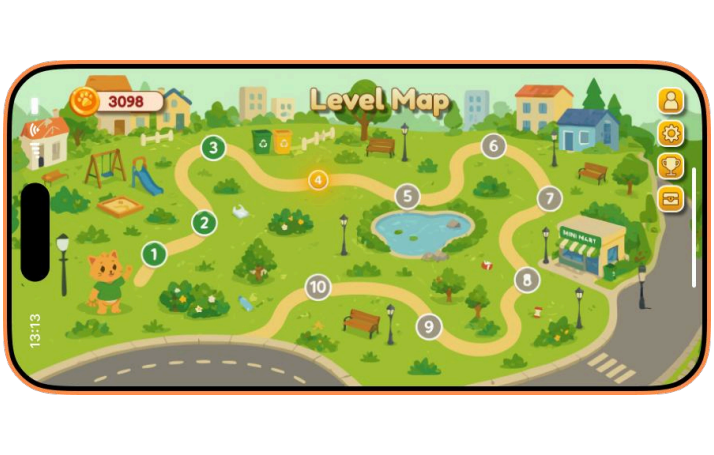
### Onboarding & Registration

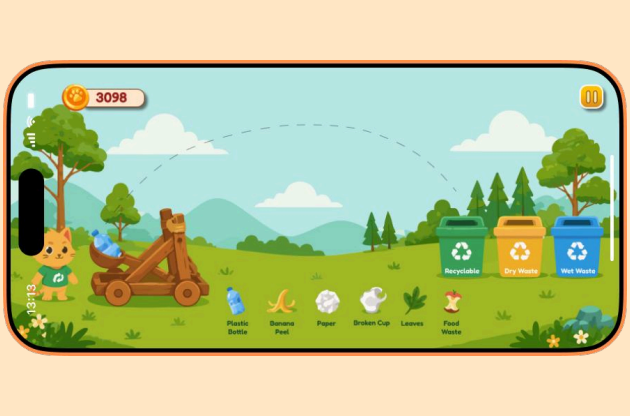
The onboarding introduces children to the game through simple visuals, interactive tutorials, and the cat mascot guiding users through waste segregation basics. Registration is kept minimal and parent-friendly, allowing quick profile creation to track progress, levels, rewards, and learning achievements.


Create Account
Enter Phone No.
Create Account
Play as Guest
How To Play
Username

### Level Map

The game uses a city-based level map where each area represents different environments such as homes, parks, schools, streets, and markets. As players progress through the city, new waste types, challenges, and segregation tasks are introduced to gradually increase learning and difficulty.

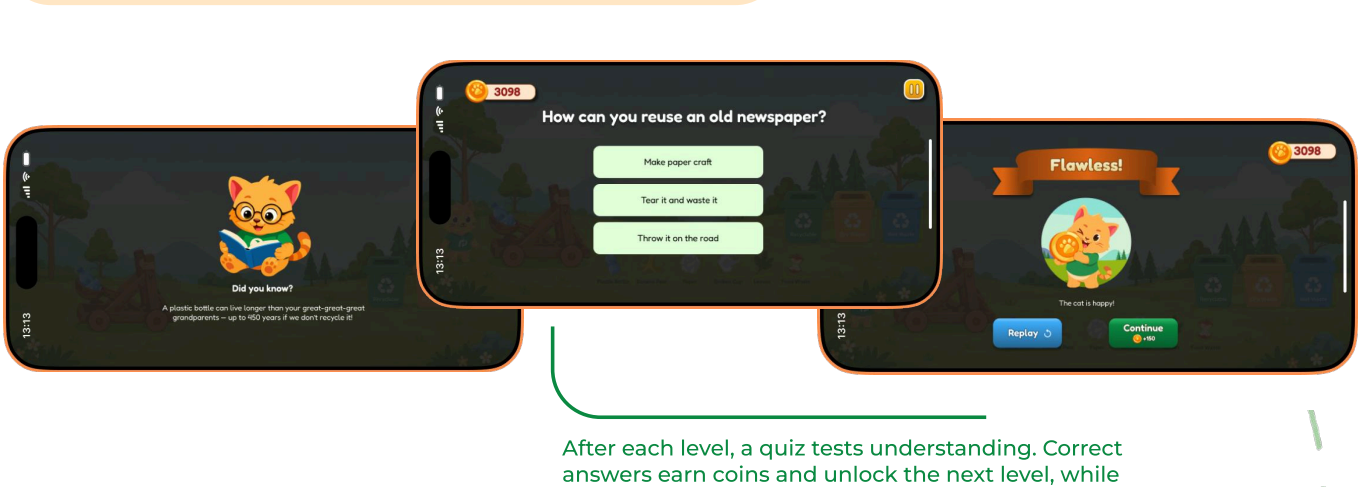

3098
Level Map


3098
Recyclable
Dry Waste
Wet Waste
Plastic Bottle
Banana Peel
Paper
Broken Cup
Leaves
Food Waste

### Learning

Children learn waste segregation through an interactive slingshot-based gameplay experience with visual and audio feedback

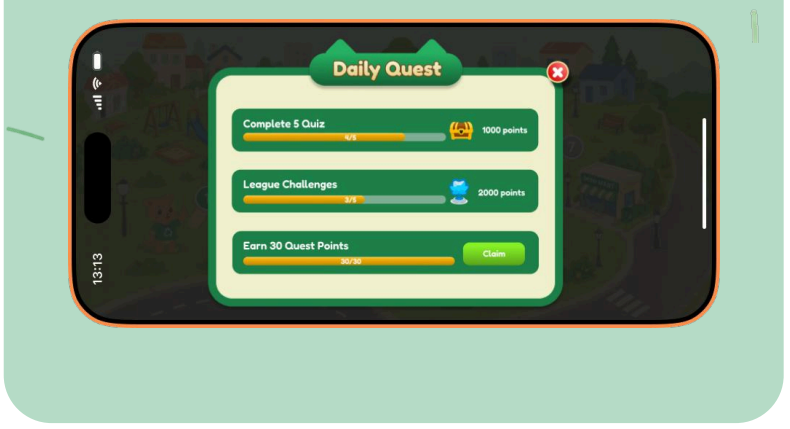
- Each level increases in complexity with new waste types, obstacles, and bin variations
- Instant visual and audio feedback reinforces learning


3098
How can you reuse an old newspaper?
Make paper craft
Tear it and waste it
Throw it on the road
Flawless!
Replay
Continue

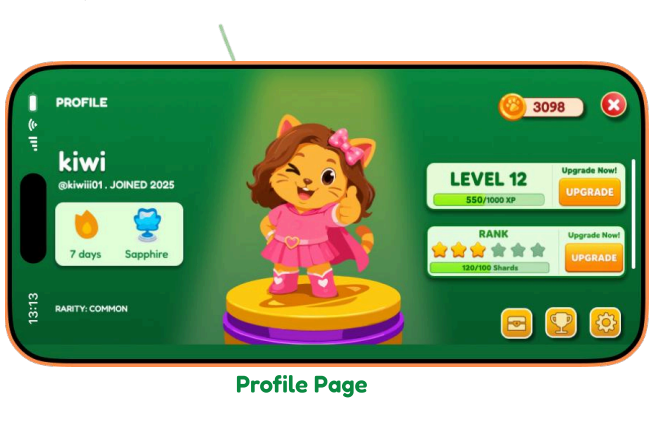
After each level, a quiz tests understanding. Correct answers earn coins and unlock the next level, while incorrect answers replay the level.

### Interactive Learning & Challenges

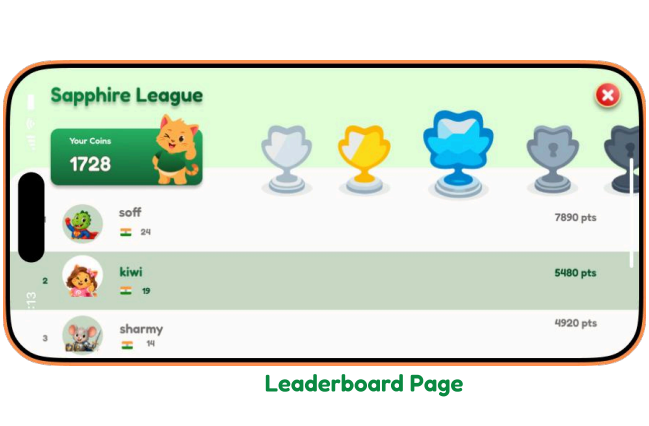
Daily waste-sorting tasks encourage consistent learning and habit-building through short interactive challenges and rewards.


Daily Quest
Complete 5 Quiz
1000 points
League Challenges
2000 points
Earn 30 Quest Points
Claim

Gamified **activities, rewards, and progression** systems make environmental education more **engaging, memorable,** and enjoyable for children.


PROFILE
3098
kiwi
@kiwiii01 . JOINED 2025
7 days
Sapphire
RARITY: COMMON
LEVEL 12
RANK
UPGRADE
UPGRADE

Profile Page


Sapphire League
Your Coins
1728
soff
7890 pts
kiwi
5480 pts
sharmy
4920 pts

Leaderboard Page

> "When learning becomes fun, and fun becomes growth."

3098
Recyclable
Dry Waste
Plastic Bottle
Banana Peel
Paper
Broken Cup
Leaves
Food Waste

## Thankyou!